

For a live connection the project file and system root folder typically is a subfolder in the Omniverse Nucleus project folder that is managed by the Omniverse Nucleus server.

Compare the examples for both use cases below.

Create and Activate a Live Connection or a Filesystem Connection

Specify these instructions to create and activate a *Filesystem Connection* with *MVA_WriteUSD*:

```
var system_root_path:string = "D:/USD_DataFiles/My_Project01/Library"  
var output_path:string = "file:///d:/USD_DataFiles/My_Project01A/  
myExport.usd"  
MVA_writeUSD(output_path+"?sysroot="+system_root_path)
```

Terminate an Open Live Connection or a Filesystem Connection

Type an empty string to terminate an open *Live Connection* or an open *Filesystem Connection* with *MVA_WriteUSD*:

```
MVA_writeUSD("")
```

Note

Make sure, that *Omniverse Composer* does not have an open connection to a previously exported simulation while starting an new export from *Plant Simulation* with the same project file and system root.

Note

For a *Nucleus Live Connection* we do not yet support the function *MVA_writeUSD()*.

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Animating the Simulation Model and Viewing the Results

Animate the Simulation Model and View the Results

Introduces you to animating your simulation model and to viewing the results of the simulation runs.

- [Animate the Simulation Model](#)
- [View and Visualize Statistics](#)
- [Switch States and Execute Actions](#)

Compare [Modeling the Material Flow](#)

Compare [Visualizing the Material Flow](#)

Compare [Getting to Know Plant Simulation](#)

Animating the Simulation Model

Animate the Simulation Model

One of your tasks when creating your simulation model will be to visually show the flow of materials through the plant you are modeling.

Plant Simulation shows the settings for the animation in 3D on the **Tab MU Animation** in the dialog **Edit 3D Properties**.

Activating and Deactivating the Animation

During the simulation run the animation shows the state of the objects and the position and the movements of the *MUs* in your simulation model.

Note

As the animation of the parts and of the icons of the object slows down the simulation, only activate it, when you actually want to show the simulation model to somebody or if you want to check if the model behaves correctly. For your overnight batch runs we recommend to deactivate it.

You can activate or deactivate **MU and State Animation** only together, not individually.

- When you activate **MUs and States** , you immediately see the flow of the parts and the blocked stations, where the *MUs* pile up.
- When you deactivate **MUs and States** while an object intends to change its state, *Plant Simulation* delays its display on screen until you activate the animation again or until the object is selected.

To deactivate **MUs and States**, click  on the **Home** ribbon tab.

To run the simulation with the setting you selected, click **Start/Stop Simulation** in the *EventController*.

Note

Clicking  on the **Home** ribbon tab activates **MU and State Animation**. To start the simulation without animating objects and *MUs*, click .

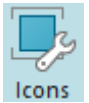
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Working with Object Icons

Work with Object Icons

We provide the *Icon Editor* to edit icons for backward compatibility.

In previous versions the 2D visualization of *Plant Simulation* used the icons of an object to display it in the *Class Library*, in the *Toolbox*, and in the *Frame*. The icons could also display the state of operation of the object, such as *failed*, *paused*, *unplanned*, etc.



You can still modify icons in simulation models, which were created in previous versions, by clicking [Edit Icons \[Home ribbon\]](#) to open the [Icon Editor](#).

The 3D visualization of *Plant Simulation 2606* does not use icons any more.

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Create an Icon

You can create a new icon and to then open an existing drawing that you would like to use.

Do one of the following:

- Click **New Icon**  on the **Edit** ribbon tab. If the drawing you open is larger than the default icon size of 41 by 41 pixels, enter another width and/or height.
- Click **Import > Import Bitmap File/Import Icon Resource**, navigate to the folder that contains the drawing you would like to use as an icon, select the file and click **Open**.
- Drag a graphics file (.gif, .bmp, .ppm, .ppm raw, .dxf or .dwg) from the Windows Explorer, etc. over the drawing window and drop it there.
- Enter a meaningful **Name** for the icon.
- To draw a line or a shape, select a drawing color and one of the drawing tools  on the **Edit** ribbon tab, and draw the shape in the drawing window.
- To change the color
 - of one or several pixels, select a color in the color palette and click the pixel(s) whose color you would like to change with one of the drawing tools.

- of a contiguous color area, select a color in the color palette and click the area whose color you would like to change with .
- To draw with two colors, you can assign a color each to the left mouse button and the right mouse button:
 - Click a color in the color palette with the left mouse button to add this color to the top drawing color field. Or
 - Double-click the color field and define a color of your choice in the dialog **Colors**.
 - Then, click the respective mouse button to draw with that color.
- You can:
 - make one or several pixels transparent, by clicking the transparency field and click the pixel(s), which you would like to make transparent with one of the drawing tools.
 - make a contiguous color area transparent, by clicking the transparency field and click the area, which you would like to make transparent with .

Note

The background of the *Frame* shines through any area you marked as transparent.

- Click  to apply the changes you made and close the *Icon Editor*.

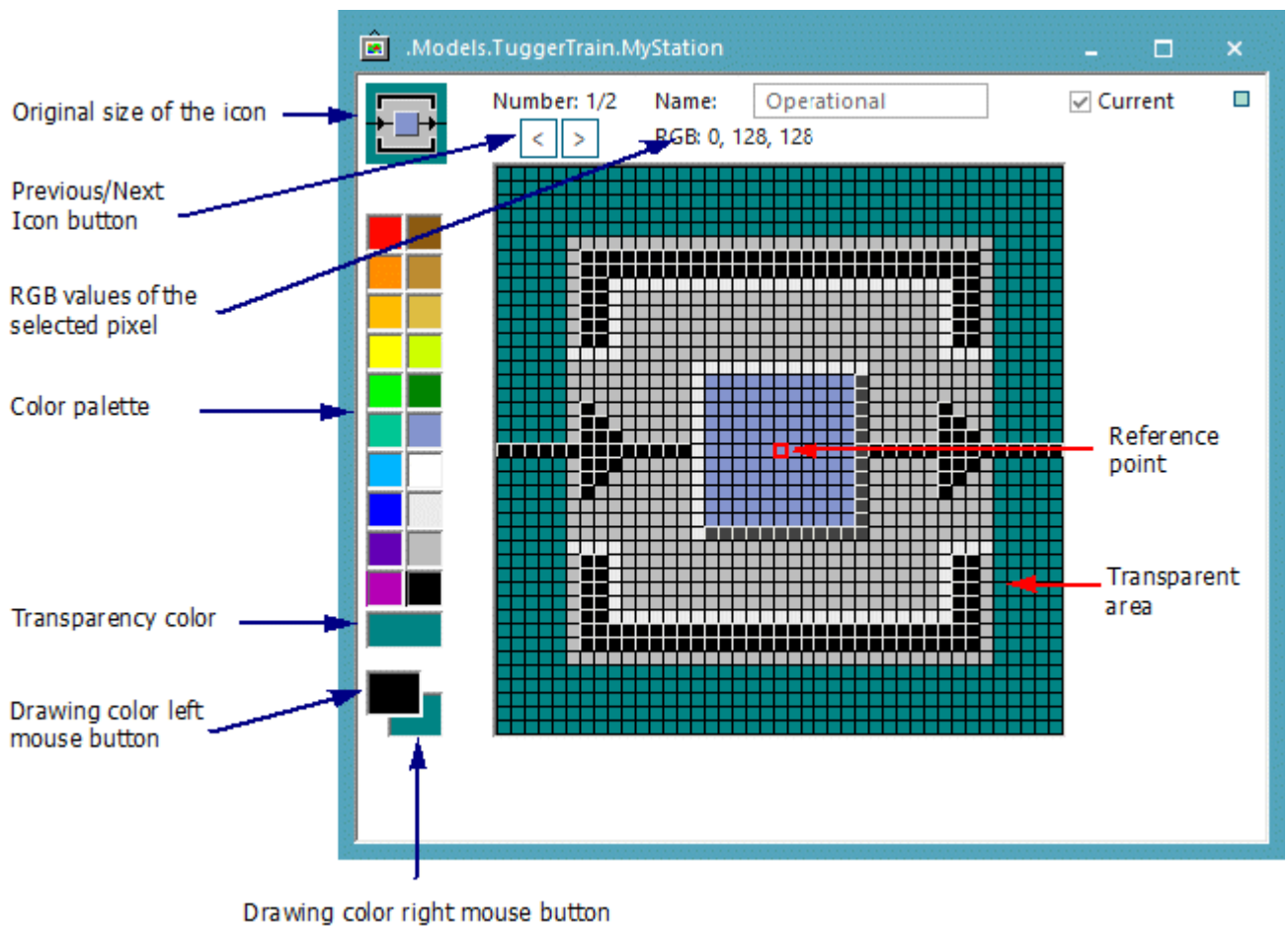
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Edit an Icon

You can edit an icon of an object.

Do one of the following:

- Right-click the object in the *Frame* and select **Edit Icons**.
- Select the object in the *Frame* and click **Edit Icons** on the **Home** ribbon tab.
- Right-click the object in the *Class Library* and select **Edit Icons**.
- Click  on the **Home** ribbon tab to edit the icons of the selected *Frame*.
- Click  on the **Home** ribbon tab to edit the icons of the selected *object*.
- Then, use the functions on the **Edit Ribbon Tab** of the *Icon Editor* to actually edit the icon.
- Click  to apply the changes you made.



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Make Areas of an Icon Transparent

You can make a contiguous area of a graphics file, which you opened, transparent, so that the background color of the *Frame* shines through.

Proceed as follows:

- Open the *Icon Editor* by clicking **Edit Icons** on the **Home** ribbon tab.
- Click  or  to navigate to the icon you want to make transparent.
- Click the color picker  and click in the background of the icon. This makes that color the active drawing color, which we want to replace.
- Select the transparency color that replaces the active drawing color in the *Color Palette* proper, not in the transparency color field!